

Amirhossein Nakhaei

Computational Social Science

Potsdam, Germany
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Research Interests

Computational analysis of culture at scale: text mining and machine learning applied to large corpora, the measurement of cultural change over time, and the design of research infrastructure that turns messy archival collections into analysable datasets.

Education

RWTH Aachen University

07/2021 – 12/2023

M.Sc. Computational Social Science, magna cum laude

Aachen, Germany

- **Thesis:** *Latent Experiments*, a method for systematically identifying naturally occurring experiments within large observational datasets. **Advisors:** Prof. Dr. Jana Lasser (then Interim Professor for Computational Social Sciences and Humanities, RWTH Aachen; now University of Graz), Dr. Mark E. Whiting (University of Pennsylvania)

Kharazmi University

09/2012 – 05/2017

B.Sc. Software Engineering, cum laude

Tehran, Iran

- **Thesis:** Farsi text super-resolution. **Advisor:** Jamshid Shanbehzadeh

Research Experience

Atlas: a platform for research cartography

07/2024 – Present

Computational Social Science Lab

University of Pennsylvania

- Designed and built a platform that integrates findings across a large collection of research literature into a navigable map of knowledge
- Applied large language models to extract, structure, and relate claims from unstructured scholarly text at scale
- atlas.seas.upenn.edu

Latent Experiments

12/2023 – 2024

M.Sc. thesis, Computational Social Science

RWTH Aachen University

- Developed a method to systematically detect latent experimental structure within existing large datasets, and benchmarked it against established causal-inference techniques
- Quantitative research design, statistical modelling, and reproducible analysis pipelines in Python

Quantifying Common Sense

02/2023 – Present

Computational Social Science Lab

University of Pennsylvania

- Technical lead on an empirical infrastructure for measuring common sense in human populations
- Designed a system supporting millions of participants across multiple languages with online adaptive sampling
- commonsense.seas.upenn.edu

Is Luck Random?

09/2022 – Present

Computational Social Science Lab

University of Pennsylvania

- Operationalised and measured “luck” in the online word game Wordle, testing hypotheses about skill versus chance in mass participation

- Assembled, cleaned, and analysed a corpus of 15 million tweets alongside directly contributed player data

Team Performance with Relational Hyper-event Models

10/2022 – 03/2023

Chair of Computational Social Sciences and Humanities

RWTH Aachen University

- Applied relational hyper-event models to team interaction data, extending a statistical network method to a new empirical domain
- Statistical analysis of roughly 100,000 match records from European football leagues, the NBA, and the NFL

Conference Presentations

Systematic preferences among causal explanations of spurious correlations	2026
<i>S. Zhang, A. Nakhaei, M. E. Whiting. IC2S2 2026</i>	<i>Poster</i>
Research Cartography with Atlas	2025
<i>M. E. Whiting, L. Gandhi, A. Nakhaei, D. J. Watts. IC2S2 2025, Norrköping</i>	<i>Tutorial/Workshop</i>
Common sense around the world	2025
<i>M. E. Whiting, A. Nakhaei, J. Nguyen, D. J. Watts. IC2S2 2025, Norrköping</i>	<i>Talk</i>
Latent experiments: learning from experiment-like situations in data	2024
<i>A. Nakhaei, M. E. Whiting. IC2S2 2024, Philadelphia</i>	<i>Talk</i>
Atlas: building a map of commensurable knowledge from existing research	2024
<i>A. Nakhaei, D. J. Watts, L. Gandhi, M. E. Whiting. IC2S2 2024, Philadelphia</i>	<i>Poster</i>
Is luck random?	2023
<i>M. E. Whiting, A. Nakhaei. IC2S2 2023, Copenhagen</i>	<i>Poster</i>

Technical & Professional Experience

Software Engineer	01/2024 – Present
<i>Systems Lab 21</i>	<i>Potsdam, Germany</i>
• Engineered scalable full-stack systems for railway testing infrastructure (EULYNX object controllers, digital interlockings) • Led the architectural redesign of the intermediate block signalling software	
Research Software Developer (working student)	04/2022 – 12/2023
<i>AIXperiment Lab & E.ON Energy Research Center, RWTH Aachen</i>	<i>Aachen, Germany</i>
• Analysed experimental economics studies using pandas and statsmodels; supported study design and implementation • Lead developer of a secure API for the EU OneNet project, supporting real-time simulation and data exchange between European power vendors	
Research Assistant, Full-stack Developer	05/2022 – 11/2022
<i>Chair of Operations Research, RWTH Aachen</i>	<i>Aachen, Germany</i>
• Built the web interface, navigation, and search tools for the GCG library of optimisation instances	
Full-stack Developer	02/2018 – 09/2020
<i>Snapp!</i>	<i>Tehran, Iran</i>
• Developed dynamic web applications (PHP, Laravel, OctoberCMS, WordPress) at one of Iran's largest technology companies	

Languages

English: fluent, IELTS 8.0/9.0; language of instruction for M.Sc. and working language of all research collaborations and conference presentations

Persian: native

German: B1

Referees

Prof. Dr. Jana Lasser

Professor for Data Analysis, IDea_Lab, University of Graz

Head, Complex Social & Computational Systems group; Associate Faculty, Complexity Science Hub Vienna

jana.lasser@uni-graz.at

Relationship: M.Sc. thesis supervisor (RWTH Aachen)

Dr. Mark E. Whiting

Chief Scientist, Pareto; Research Fellow, Computational Social Science Lab, University of Pennsylvania

markew@seas.upenn.edu, whiting.me

Relationship: M.Sc. thesis co-supervisor; research supervisor at CSSLab (2022–present)